PSI Center for Scientific Computing, Theory and Data

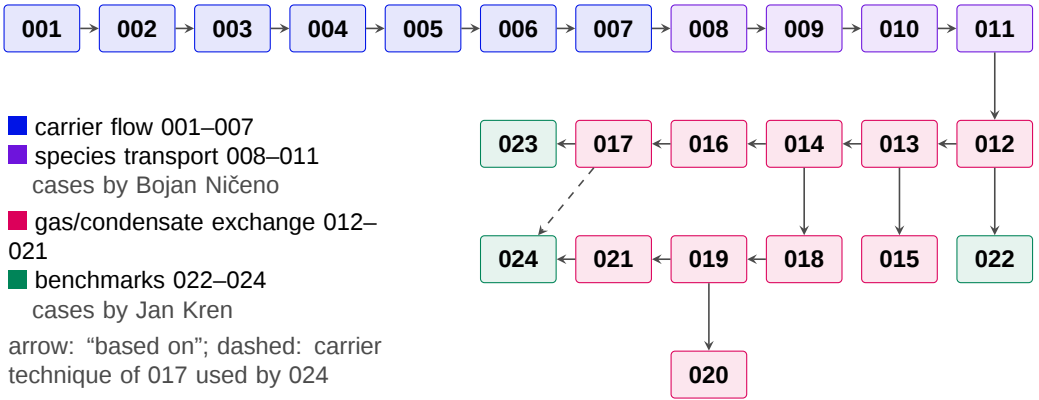
The LESTO case ladder

From a laminar pipe to the benchmark studies, one change per case

Jan Kren

LESTO Task 3.4 | Villigen PSI, 15 October 2026

The ladder: 24 cases, one change each



carrier flow 001–007

species transport 008–011

cases by Bojan Ničeno

gas/condensate exchange 012–021

benchmarks 022–024

cases by Jan Kren

arrow: “based on”; dashed: carrier technique of 017 used by 024

Every case: the same four parts

readme.md of each case

- “This case is based on case N”
- the changed dictionary lines
- the numbers to check
- how to run it

001–011 (Bojan Ničeno)

Run in place: `gmsh`, `simpleFoam` or `rhoSimpleFoam`; 008–011 read the mesh of 001 and the carrier of 007 at 261 through links.

012–024 (Jan Kren)

- `./Allrun`: the default check into `work/`
- ends with one line PASS, FAIL or NOTRUN
- compared with `out_excerpt` or `expect.json`
- `./Allrun-ladder 012-024`: one line per case

Figures in this deck

Real outputs: tracked logs and records, or runs made for this deck in a scratch copy of `run/` on 3 cores (`runs/runs.json`).

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1 Carrier flow 001–007

cases by Bojan Ničeno

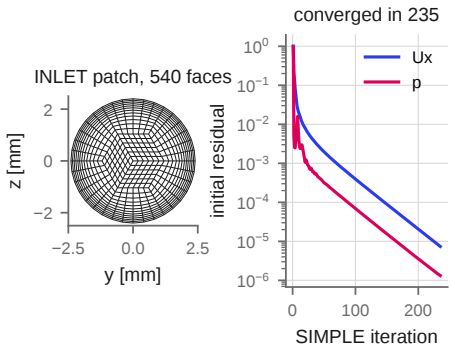
001 first-pipe-flow-simpleFoam

The first case | Bojan Ničeno

Laminar helium pipe, gmsh mesh, steady `simpleFoam`

```
INLET  uniform (0.095 0 0); // m/s
nu     1.6265e-4;   // m2/s, He
```

- R = 2.4 mm, L = 0.69 m, 228,960 hexahedra
- `checkMesh`: Mesh OK
- converged in 235 SIMPLE iterations



Mesh of `pipe.msh`; `001/out`

Run: `gmsh`, `gmshToFoam`, `simpleFoam`

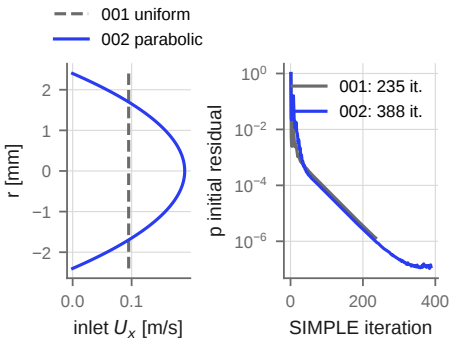
002 parabolic-inlet

Based on case 001 | Bojan Ničeno

Parabolic inlet profile, coded`FixedValue` in `0/U`

```
const scalar R    = 0.0024;
const scalar Umax = 0.19;
Ux = Umax*(1.0 - r2/(R*R));
```

- bulk velocity about 0.095 m/s: same flow rate as 001
- internal field still uniform 0.095 m/s
- converged in 388 iterations (001: 235)



`0/U`; `001/out`, `002/out`

Run: as 001, `simpleFoam > out`

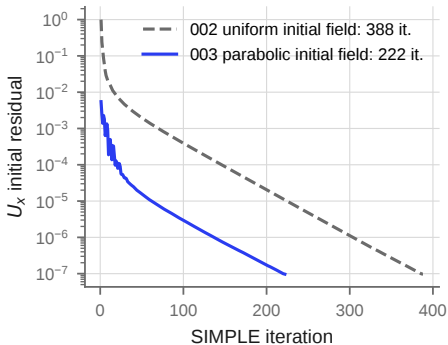
003 parabolic-initialization

Based on case 002 | Bojan Ničeno

Parabolic initial field with `setExprFields`

```
U { field U; expression #{ vector(
  0.19*(1.0 - (sqr(pos().y())
   + sqr(pos().z()))/sqr(0.0024)),
  0, 0) #}; }
```

- inlet boundary condition unchanged from 002
- converged in 222 iterations (002: 388)



`002/out`, `003/out`

Run: `setExprFields; simpleFoam`

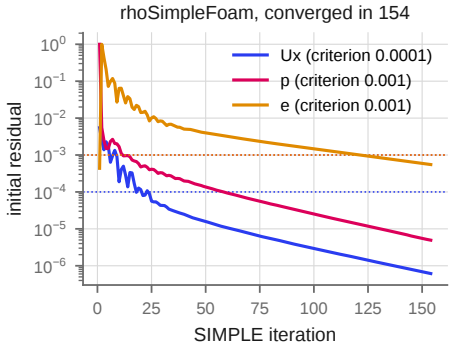
004 energy-with-rhoSimpleFoam

Based on case 003 | Bojan Ničeno

`rhoSimpleFoam` and the energy equation

```
type heRhoThermo;
equationOfState rhoConst;
Cp 5193; Pr 0.916;  // T = 300 K
```

- p in Pa: 101325 at the outlet
- criteria (`fvSolution`): U 1e-4, p 1e-3, e 1e-3
- converged in 154 iterations



`004/out`; dotted: the criteria

Run: `setExprFields; rhoSimpleFoam`

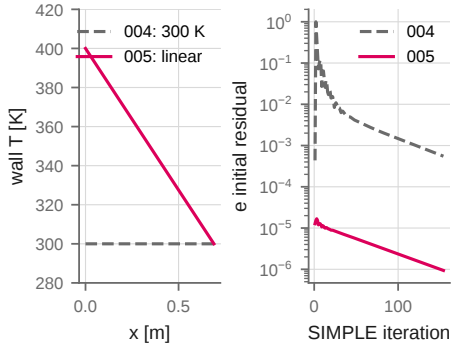
005 varying-wall-temperature

Based on case 004 | Bojan Ničeno

Linear wall temperature, 400 K to 300 K

```
WALL { type codedFixedValue;
  T = 400.0 - 100.0*x/0.69; }
INLET  uniform 400;
```

- internal T initialised with the same ramp
- energy residual far lower than in 004
- converged in 154 iterations



`0.orig/T`; `004/out`, `005/out`

Run: `setExprFields; rhoSimpleFoam`

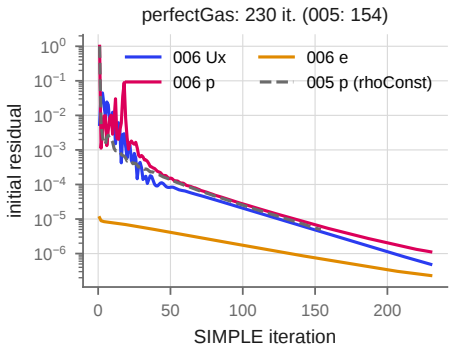
006 variable-density

Based on case 005 | Bojan Ničeno

Ideal-gas density from local p and T

```
type            hePsiThermo;
equationOfState perfectGas;
molWeight       4.0026;
```

- was `heRhoThermo + rhoConst`
- viscosity, Cp and Pr still constant
- converged in 230 iterations (005: 154)



`006/out`; dashed: `005/out`

Run: `setExprFields; rhoSimpleFoam`

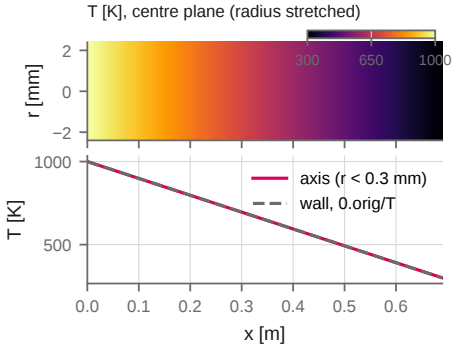
007 variable-properties-high-temperature

Based on case 006 | Bojan Ničeno

Sutherland transport, wall 1000 K to 300 K

```
transport sutherland;
As 1.4683e-06;  Ts 83.40;
T_wall = 1000.0 - 700.0*x/0.69;
```

- writes `mu` and `kappa` (coded function object)
- converged in 261 iterations: the carrier at 261
- 28 mL/min, T and U overwritten: development carrier



`007/261`, foamToVTK

Run: `setExprFields; rhoSimpleFoam`

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2 Species transport 008–011

cases by Bojan Ničeno

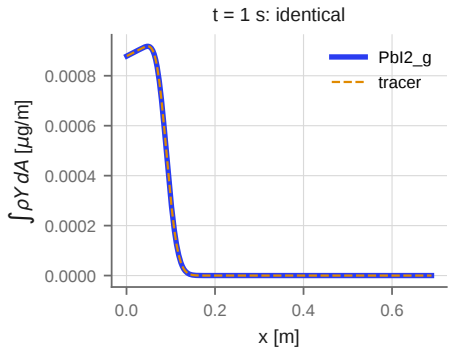
008 one-species-transport-rhoFixedFlowFoam

Based on case 007 | Bojan Ničeno

`rhoFixedFlowFoam`: gaseous species on the frozen carrier

```
species ( PbI2_g tracer );
PbI2_g { state gas; D 1.0e-5; }
tracer { state gas; D 1e-5; }
```

- no momentum, pressure or energy solve
- same D, inlet and initial field: identical Y
- run for this deck to 1 s: max $|\mathrm{PbI}_2 - \mathrm{tracer}|$/max = 0



Deck run: 1 s, 3 ranks, 257 s

Run: `rhoFixedFlowFoam > out`

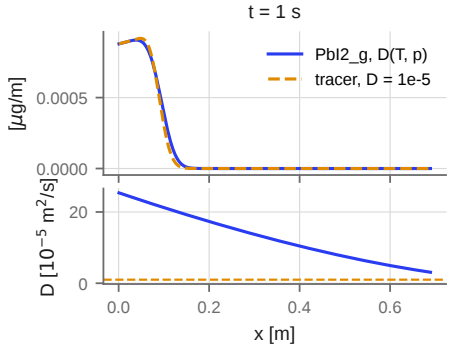
009 variable-diffusivity

Based on case 008 | Bojan Ničeno

PbI_2–He diffusivity from the carrier T and p

```
PbI2_g { diffusivityModel PbI2He; }
tracer { diffusivityModel constant;
         D 1e-5; }
```

- D evaluated once at start-up, then fixed
- `D_PbI2_g`: 3.089e-5 to 2.536e-4 m^2/s
- tracer uncoupled, solved after PbI_2



Deck run: 1 s, 3 ranks, 559 s

Run: `rhoFixedFlowFoam > out`

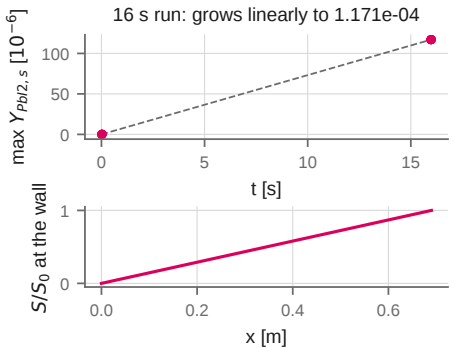
010 handles-solid-species

Based on case 009 | Bojan Ničeno

A solid species and a mock thermochemistry call

```
PbI2_s { state solid; }
S = S0*(Thot - T)/(Thot - Tcold);
S0 1e-6; Thot 1000; Tcold 300;
```

- solid: only $\partial_t(\rho Y) = S$, `diagonal` solver
- source kept in the 15,264 wall-adjacent cells
- no coupling: max Y grows linearly in time



Tracked `010/out_excerpt`, 16 s run

Run: `rhoFixedFlowFoam > out`

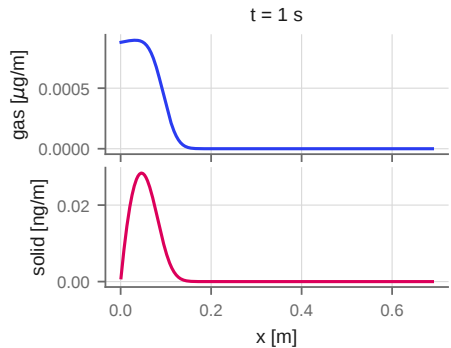
011 coupled-species-sources

Based on case 010 | Bojan Ničeno

Gas PbI_2 deposits as solid: equal and opposite sources

```
rate = K*max(Y_PbI2_g, 0)
  *max((Thot-T)/(Thot-Tcold), 0);
K 1.0;  // kg/(m3 s), WALL cells
```

- gas sink implicit; solid takes what the gas lost
- excerpt: net thermochemistry source 0 kg/s
- full mass ledger not yet (comes in 012)



Deck run: 1 s, 3 ranks, 399 s

Run: `rhoFixedFlowFoam > out`

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3 Gas/condensate exchange 012–021

cases by Jan Kren

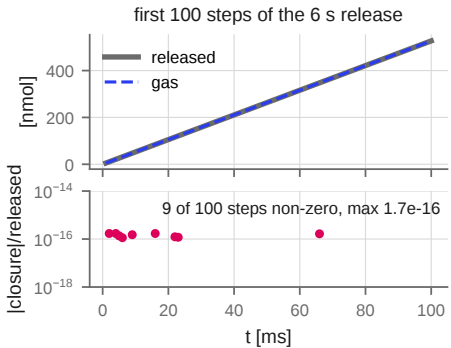
012 temperature-model

Based on case 011 | Jan Kren

Wall law of Lobresco et al. (2026), Eq. 5, a release boat and a mass ledger

```
model temperature;
temperatureCoeffs {A 220;
  k 8e-3; Tdep 680;}
boat {mode release; amount 3.16e-5;
      duration 6;}    // mol, s
```

- 15,264 wall elements, v_d = 0.0124421 m/s
- closure step 5: -1.36×10^{-16}; 6 s: 1.8e-14
- deck run: max |closure|/released 1.7e-16



Deck run: `steps 100`, 1 core, 260 s

Run: `./Allrun`: excerpt PASS

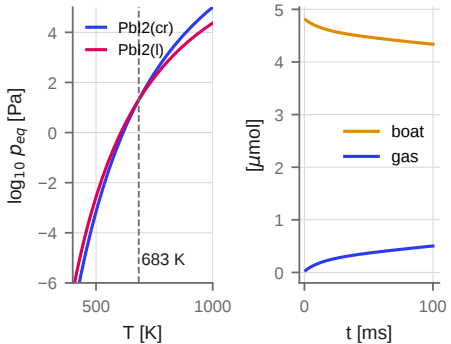
013 hks-inventory

Based on case 012 | Jan Kren

Hertz–Knudsen–Schrage law at the wall and in an evaporating boat

```
model HKS; equilibrium table;
phases ("PbI2(cr)" "PbI2(l)");
mode inventory; amount 4.84e-6;
```

- p_{eq}: lower of crystal and liquid, cross at 683 K
- $\lambda\,\Delta t$ = 1000 to 1900: stiff, implicit
- closure -1.9×10^{-16}; deck run: boat 4.84 $\rightarrow$ 4.34 μmol in 0.1 s



p_{eq} table; deck run, 100 steps

Run: `./Allrun`: excerpt PASS

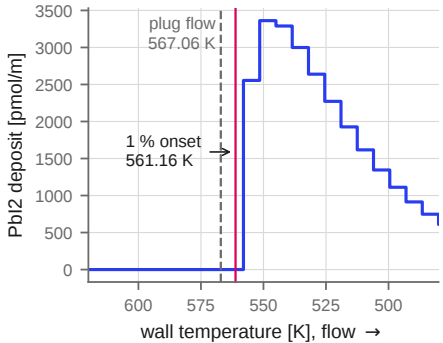
014 hks-channel

Based on case 013 | Jan Kren

The synthetic plane channel, 1000 K to 350 K: axial profiles and T_{dep}

```
// 0.1 m x 4.8 mm, 100 x 24 cells
Tin 1000; Tout 350;
massFlux 0.01743;
onset {fraction .01;}
```

- 1 % onset 561.164 K, 5.89 K below plug flow (gate 10 K)
- 99.5 % of the release on the wall at 3 s
- closures 2.5e-16 (pair), 4.05e-16 (Pb, I)



Deck run, wall profile at 3 s

Run: `./Allrun`, 10 s: PASS 9/9

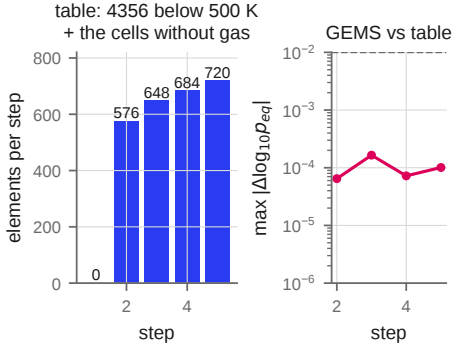
015 hks-gems-local

Based on case 013 | Jan Kren

p_{eq} from GEMS3K for every wall element and boat cell, every step

```
HKSCoeffs {equilibrium GEMS;}
GEMSCoeffs {mode local;
  minTemperature 500;
  maxLog10Deviation 0.01;}
```

- 576 to 720 calls per step, 0 failures
- GEMS and table agree to 1e-5 to 1e-4 decades
- balance and closure those of 013 to round-off



Tracked `out_excerpt`, 5 steps

Run: needs the bridge build: not run

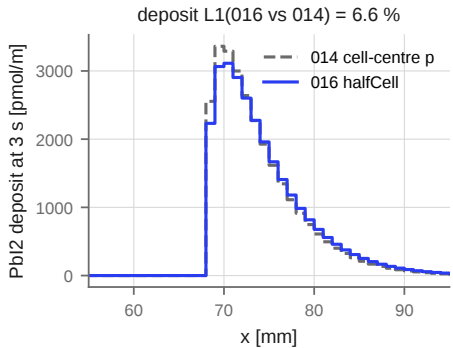
016 hks-channel-halfcell

Based on case 014 | Jan Kren

Half-cell diffusion resistance in series with the HKS law

```
interface {patches (WALL);
           wallResistance halfCell;}
```

- conductance 63.7–107 → 0.98–0.99 m/s
- onset moves 5 mK: 561.1645 → 561.1593 K
- peak 7.4 % lower, one bin downstream; L1 6.6 %



Deck runs 014 and 016, at 3 s

Run: `./Allrun`, 10 s: PASS 9/9

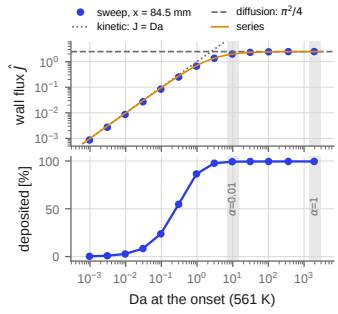
016 hks-channel-halfcell: Damköhler sweep

Case 016 unchanged except α | Jan Kren

Accommodation α over six decades: Da from 10^{-3} to 2×10^{3}

```
// PbI2_g pair and HKSCoeffs:
accommodation 1e-6 ... 1;
// 13 runs, 5 refined
```

- $\hat{J} = \mathrm{Da}$ for Da $\leq$ 0.01; plateau $\pi^2/4$ above 300
- deposited 0.3 $\rightarrow$ 99.5 %; onset stays at 561 K
- $\alpha = 0.1$: profile 1.4 %; $\alpha = 0.01$: 14 %



Record `manuscript/data/damkohler`

Run: `./Allrun-damkohler`, a few minutes

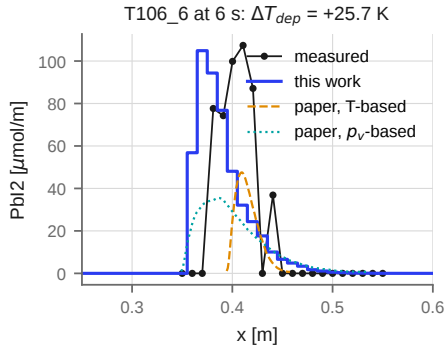
017 physical-carrier

Based on case 016 | Jan Kren

The physical carrier of Lobresco T106_6: 106 mL/min, measured wall T

```
// wedge 5 deg, R 2.4 mm, 212 x 48
massFlowRate 4.3817e-09; // kg/s
interfaceTemperature wall;
```

- carrier audit PASS, cell flux defect 1.2e-17
- T_{dep} 839.09 K: +25.73 K against 814 K
- peak 4 cm upstream; closure 2.1e-16



Record `023/cases/coarse/T106_6`

Run: `./Allrun`, about 8 min

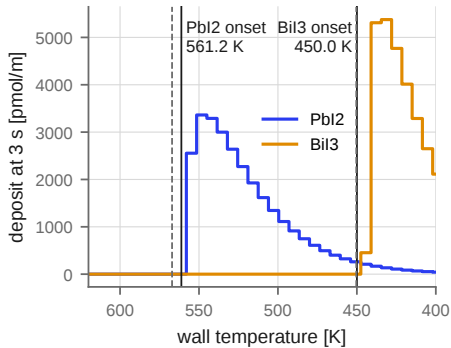
018 pbi2-bii3-channel

Based on case 014 | Jan Kren

A second, independent pair: BiI_3 from the same boat

```
BiI3_g {condensed BiI3_s;
  HKS {accommodation 1; ...}
  vapourPressure {phases
    ("BiI3(cr)" "BiI3(l)");}}
```

- onsets: PbI_2 561.164 K, BiI_3 449.979 K
- plug flow: 567.057 K and 450.383 K
- closures Pb 4.05e-16, Bi 4.56e-16, I 4.37e-16



Deck run, wall profiles at 3 s

Run: `./Allrun`: PASS 8/8

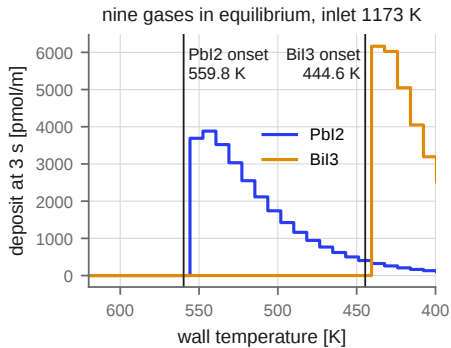
019 pbbi-gas-speciation

Based on case 018 | Jan Kren

Nine gases in homogeneous equilibrium, every cell and step

```
respeciation {engine kernel;
  species (Pb_g PbI_g PbI2_g Bi_g
    Bi2_g BiI_g BiI3_g I_g I2_g);
  minTemperature 350;}
```

- inlet 1173 K: onsets 559.778 K and 444.563 K
- vs 018 at 1173 K: 0.0000844 % and 0.1332 %
- element closures 4.3–5.0e-16



Deck run, wall profiles at 3 s

Run: `./Allrun`, 457 s: PASS 7/7

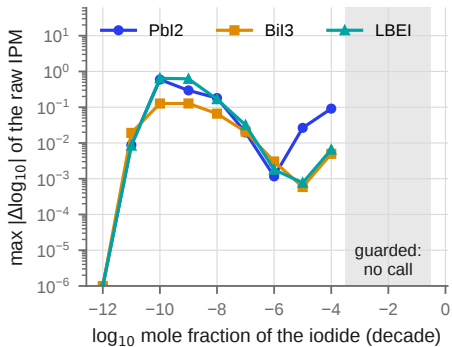
020 pbbi-gems-speciation

Based on case 019 | Jan Kren

Re-speciation by GEMS3K: modes `frozen` and `local`

```
respeciation {engine GEMS;
  GEMS {mode frozen;
    carrier "He(g)";
    minMoleFraction 1e-12;}}
```

- L1 frozen vs kernel: 1.03e-10, 1.06e-7
- local vs frozen: 0; closure 6.7e-16
- 132.2 μs per local call (5,814,420 calls)



`020/qualification.json`

Run: needs the bridge build: not run

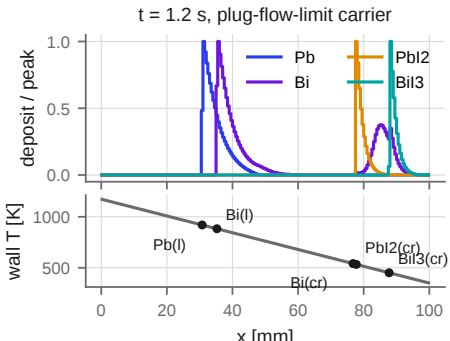
021 lbe-metal-vapour

Based on case 019 | Jan Kren

Metal vapour, shared condensates and reactive wall channels

```
BiI {reactant BiI_g;
  condensed Bi_s; reversible no;
  stoichiometry {BiI_g -3;
    Bi_s 2; BiI3_g 1;}}
```

- five onsets within 3.89 K of plug flow
- half step changes none of them
- closures 3.7–4.6e-16



Deck run, profiles at 1.2 s

Run: `./Allrun`, 45 s: PASS 8/8

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4 Benchmarks 022–024

cases by Jan Kren

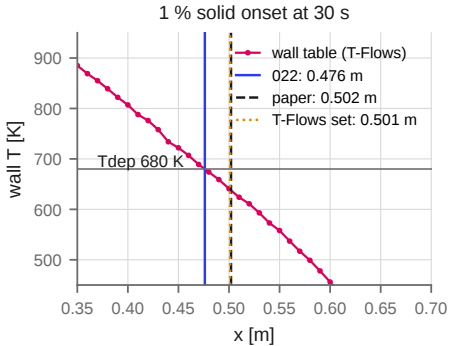
022 paper-temperature-model

Based on case 012 | Jan Kren

Paper mode: code-to-code with T-Flows, Figs. 8–10 and Table 2

```
// setFrozenCarrier: rho 1,
//   1.67e-6 m3/s (100 mL/min)
interface {layer distance;
           distance 1e-4;}
amount 3.16e-5; duration 15;
```

- gas plateau at 16 s: +0.5 % against Fig. 9
- solid onset -25.9 mm; T-Flows set: -0.9 mm
- Table 2: all ten lines within max(2 K, ΔT_{bin})



Wall table; onsets: readme, notes

Run: `./Allrun`, 38 s: excerpt PASS

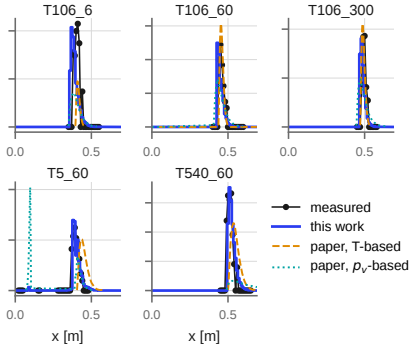
023 lobresco2026

Based on case 017 | Jan Kren

All five PbI_2 experiments of Lobresco et al. (2026), two meshes

```
// T106_6 T106_60 T106_300
// T5_60 T540_60
prod:   318 x 72 = 22,896 cells
coarse: 212 x 48 (case 017)
```

- ΔT_{dep}: +26, +19, +16, -11, -14 K
- RMS 18 K; the paper's two models: 35 K
- coarse = prod within 0.3 K; closure 3.3e-16



Records `cases/prod`; data CC BY 4.0

Run: `./Allrun prod T106_6 ...`

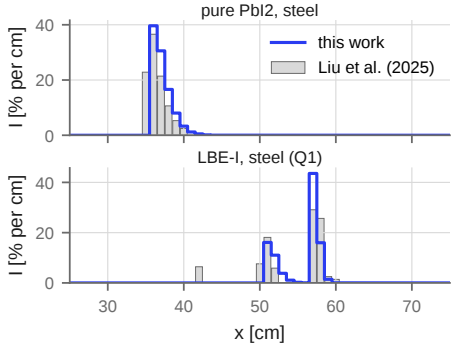
024 liu2025

Based on case 021 (+ carrier of 017) | Jan Kren

Liu et al. (2025): Pb–Bi–I chemistry on physical carriers

```
interfaceTemperature wall;
wallResistance halfCell;
// Q1; Q2 at 0.1, 1, 10 % saturation
```

- 576 x 72, limitedLinear: 19 PASS
- PbI_2 peak 36 cm / 366.5 °C vs 368 ± 18 °C
- mesh 1.98 % (one case); correctors ≤ 0.021 %



`production_576x72_limitedLinear`

Run: `./Allrun full` (6 h per case)

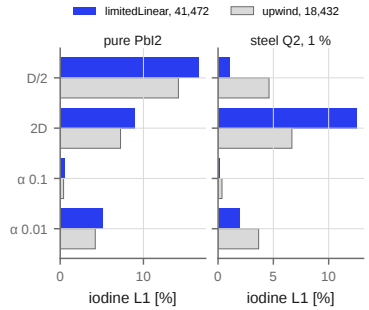
024 liu2025: sensitivities, mesh and GEMS3K

Production cases of 024 | Jan Kren

The checks behind the production runs, made after the review

```
production_sensitivities/   // 8 runs
scheme_mesh_study/upwind_L4/ // 4 runs
```

- limitedLinear: $\alpha \leq$ 5.14 %, D 1.07–16.65 %
- 41,472 $\rightarrow$ 93,312 cells, upwind: 2.13–7.30 %
- GEMS3K in every cell: profiles within 2.1e-4



Record `024/production_sensitivities`

Run: `make_cases.py`; 17–28 h per case

What the ladder demonstrates

One change per case, each checked

- carrier: pipe to Sutherland helium (001–007)
- transport: frozen carrier, species list, coupled sources (008–011)
- exchange: every ledger closes at round-off
- onsets within 3.89 K of plug flow (021)
- wall law: kinetic and diffusion limits (016 sweep)
- experiments: T_{dep} RMS 18 K (023)

Where it is used

- methods paper: 016 sweep, 023, 024
- deliverable D38: the case ladder of its availability section
- `./Allrun-ladder 012-024`: regression of every check

Deck runs, 15.10.2026

PASS: 012–014, 016, 018, 019, 021, 022. To 1 s: 008, 009, 011. The rest: tracked logs and records.